BIASBUSTERS: UNCOVERING AND MITIGATING TOOL SELECTION BIAS IN LARGE LANGUAGE MODELS

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ABSTRACT

Agents backed by large language models (LLMs) often rely on external tools drawn from marketplaces where multiple providers offer functionally equivalent options. This raises a critical point concerning fairness: *if selection is systematically biased, it can degrade user experience and distort competition by privileging some providers over others*. We introduce a benchmark of diverse tool categories, each containing multiple functionally equivalent tools, to evaluate tool-selection bias. Using this benchmark, we test seven models and show that unfairness exists with models either fixating on a single provider or disproportionately preferring earlier-listed tools in context. To investigate the origins of this bias, we conduct controlled experiments examining tool features, metadata (name, description, parameters), and pre-training exposure. We find that: (1) semantic alignment between queries and metadata is the strongest predictor of choice; (2) perturbing descriptions significantly shifts selections; and (3) repeated pre-training exposure to a single endpoint amplifies bias. Finally, we propose a lightweight mitigation that first filters the candidate tools to a relevant subset and then samples uniformly, reducing bias while preserving good task coverage. Our findings highlight tool-selection bias as a key obstacle for the fair deployment of tool-augmented LLMs.

1 INTRODUCTION

Large language models (LLMs) have transformed natural language processing, achieving near-human performance on tasks ranging from code generation to creative writing (Naveed et al., 2024; Luo et al., 2024). Yet LLMs cannot directly act in the world: they cannot query databases, fetch live information, or invoke external services. Additionally, their knowledge remains frozen at training time, leaving them prone to "hallucinations" when asked about events beyond their cutoff (Ji et al., 2023). Augmenting LLMs with external "tools" / APIs addresses these shortcomings by allowing models to delegate specialized functions to dedicated services (Qu et al., 2025). It endows LLMs with the ability to act, a core capability often associated with LLM *agents* (Chowa et al., 2025). A crucial step within the typical tool-usage pipeline is the multi-stage tool-selection process: given an instruction to the LLM, (i) retrieve a short list of the most relevant candidate tools based on the user query (with, e.g., highest semantic similarity) from a potentially large database of tools, (ii) insert their metadata into the prompt, (iii) have the LLM reason and pick one to solve (one of) the necessary user task(s). However, this process introduces a new challenge: bias (see Figure 1). An LLM may prefer certain tools not for their relevance or accuracy, but because of superficial metadata, i.e., tool names, descriptions, or prompt ordering. Such bias can degrade user experience by repeatedly selecting slow or unreliable services and can, therefore, also inflate operational costs. Additionally, under pay-per-request pricing for tool use, consider the scenario where such biases are systematic across frontier LLMs: if they consistently favor tools from a single provider, this creates major market unfairness, disadvantaging competing providers offering functionally equivalent services (RapidAPI, 2025a; Economize, 2024).

In this paper, we make three key contributions. **First**, we present a large-scale benchmark for measuring tool selection bias, which we use to conduct the first empirical study of tool-selection bias in LLMs. We introduce total variation–based metrics to quantify selection imbalances, assemble a

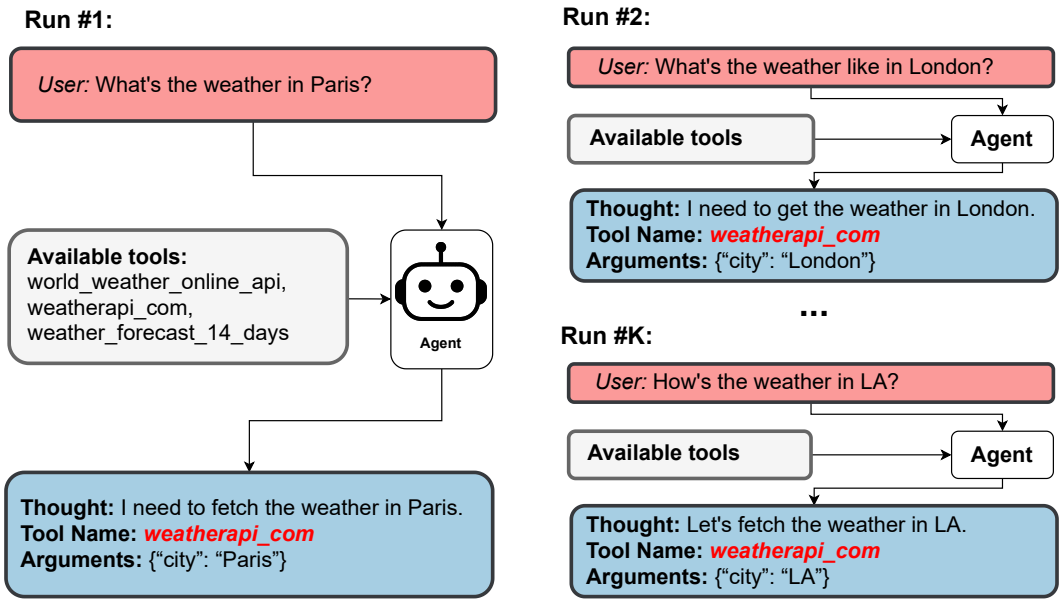

Figure 1: Tool-calling enables LLMs to act through external services, but the selection process introduces bias. Models may favor certain tools based on superficial metadata or position rather than relevance (here "weatherapi_com" is preferred), leading to a potential degraded user experience and unfair concentration of calls. If such biases are systematic across frontier LLMs, they risk distorting entire tool marketplaces, disadvantaging functionally equivalent competitors.

novel benchmark of clusters of tools with equivalent functionality, and conduct extensive experiments across multiple models and configurations. These tell us that bias exists to a certain extent for *all* models tested; models either fixate on a single provider or disproportionately prefer earlier-listed tools in context. **Second**, after confirming the existence of bias, we investigate the root causes of these biases. We define a rich set of tool-level features and use regression and permutation-importance analyses to identify which factors predict selection. We further perturb metadata fields to test their influence and probe whether continued pre-training on biased data can induce persistent preferences. Our findings show that semantic alignment between query and tool description is the strongest interpretable driver of tool choice, metadata interventions on descriptions reliably steer choices, and biased pre-training amplifies preferences. However, none of these fully explain the behavior. **Third**, we propose a lightweight mitigation strategy: filtering the supplied tools to a subset of relevant ones and sampling uniformly among them. This approach substantially reduces bias while still ensuring the user task can be solved.

By uncovering, explaining, and addressing tool-selection bias, our work illuminates a critical blind spot in tool-augmented LLM research. Our contributions go beyond diagnosis: we provide reproducible resources and a simple mitigation strategy that practitioners can adopt immediately. More broadly, we aim to set a foundation for fairer, more reliable tool-calling systems and a precedent to judge tool-calling LLM applications not only by their accuracy but also by the equity of their interactions with external ecosystems. All resources are available online[1].

2 Related Work

Fairness & Bias in LLMs. Bias in large language models emerges from their opaque training on vast web-scale data, often carrying forward and amplifying human prejudices and reasoning shortcuts (Gallegos et al., 2024). Much of the literature has focused on identifying and mitigating these undesirable behaviors (Schick et al., 2021; Gallegos et al., 2024; Bouchard, 2024). Unwanted behaviors in LLMs manifest as social bias, where stereotypes shape outputs and lead to unequal treatment of protected groups, or as cognitive bias, reflecting human-like shortcuts such as anchoring or confir-

[1] All resources are available on the public GitHub repository.

mation effects (Lou & Sun, 2024; Itzhak et al., 2024; O'Leary, 2025). Measuring these phenomena requires domain-specific metrics. For example, to measure how likely a model is to associate a neutral attribute with one group over another, one can utilize the normalized log-probability bias score (Kurita et al., 2019). Mitigation strategies are diverse and can be deployed at multiple stages (e.g., at pre-/post-inference via augmentation of training data or rewriting of outputs or during training via fairness-aware objectives) (Gallegos et al., 2024). However, no single approach suffices for every scenario; effective debiasing requires interventions matched to the task's fairness goals and deployment context. We are not aware of prior work on bias in the tool-use paradigm.

Fairness in Tool Selection. Research on tool-selection bias is limited; however, some works in information retrieval address related issues such as positional or ranking bias (Dai et al., 2024; Ziems et al., 2024). Similarly, in code generation, favoritism for certain providers has been observed (Zhang et al., 2025). However, these studies do not examine the specific setting of tool selection among providers. Recent studies do show that LLMs can develop systematic preferences in this setting, for example, for selecting tools with manipulated metadata (Mo et al., 2025; Faghih et al., 2025). Analogously, one study shows an advanced attack on tool selection which splits malicious tool entries into fragments optimized for retrieval and selection, achieving high success rates while evading defenses (Shi et al., 2025). These findings reveal that tool-selection mechanisms are highly fragile and susceptible to both naïve and advanced exploitation. In contrast to overt adversarial attacks, our study focuses on subtler sources of bias: small, non-adversarial differences in phrasing or metadata that can still distort fairness in tool selection.

3 BIASBUSTERS: HOW WE UNCOVER, EXPLAIN, AND MITIGATE BIAS.

In this section, we present our end-to-end framework for uncovering and explaining tool-selection bias in LLMs. We start in Section 3.1 by introducing our formal definition of this bias. In Section 3.2, we describe how we generate a comprehensive benchmark of API clusters and user queries that enables systematic measurement of selection behavior. Section 3.3 details our analysis pipeline for explaining bias.

3.1 BIAS DEFINITION

Tool-selection bias captures the extent to which an LLM systematically favors certain APIs over others for reasons unrelated to their true utility. Formally:

Definition 3.1 *Tool-selection bias is the systematic tendency of a model to favor certain APIs over others for reasons unrelated to the APIs' true relevance or utility for the task.*

To quantify this bias, we compare a model's empirical selection rates against an ideal uniform choice. Consider a cluster of K APIs that can all solve a given query q. A perfectly unbiased model would select each API with probability $1/K$, forming a uniform distribution U. Let $P^{\mathrm{API}} = (P_1^{\mathrm{API}}, \ldots, P_K^{\mathrm{API}})$ denote the expected empirical selection-rate distribution over the cluster under all orderings of the API list. We measure cluster-level bias via the total variation distance

$$\delta_{\mathrm{API}} \;=\; \tfrac{1}{2}\sum_{i=1}^{K}\Big|P_i^{\mathrm{API}} - \tfrac{1}{K}\Big| \;=\; \mathrm{TV}\big(P^{\mathrm{API}},\, U\big).$$

Prior work has shown that list order itself can introduce bias: when multiple identical tools appear in sequence, the first position is favored Faghih et al. (2025). To capture this effect, we also record the expected selection-rate distribution over absolute positions, P^{pos}, and compute $\delta_{\mathrm{pos}} = \frac{1}{2}\sum_{i=1}^{K}\Big|P_i^{\mathrm{pos}} - \frac{1}{K}\Big|$. Combining both yields an overall bias metric: $\delta_{\mathrm{model}} = \frac{\delta_{\mathrm{API}}+\delta_{\mathrm{pos}}}{2}$. Note that a high positional bias δ_{pos} can be mitigated by randomizing API order at prompt time, whereas API-level bias δ_{API} requires deeper intervention.

3.2 DATASET GENERATION

We build on the ToolLLM pipeline (Qin et al., 2024), which provides a large repository of APIs scraped from RapidAPI (RapidAPI, 2025b). To measure tool-selection bias, we construct a benchmark of functionally interchangeable APIs. Specifically, we cluster APIs into groups performing